

canyon-0.1.1 Manual

C. Naaktgeboren
<https://github.com/cnaak/canyon.typ>
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1 Canyon's End User in Full Control Approach

Mainstream Typst templates are written so as to bring functionality and to be used in an `#import` then `#show workflow`; however, that frequently comes at the cost of introducing (i) new syntax and (ii) opaque template-side code and settings.

Any template end user that had the need to customize things not anticipated or provided for by the template's author have experienced this shortcoming.

Canyon (canyon-0.1.1) explores a different approach in which a new canyon-based document is a book *stencil* that users can freely edit and adapt, with:

- *No* opaque template-side code or settings;
- *No* canyon `#import` statements;
- *No* canyon-specialized `#show` rules;
- *Full access* to the entire Typst project;
- The *familiar* Typst syntax to work with; and
- Full *separation of concepts* for (i) metadata, (ii) configuration, (iii) elements, (iv) layout, and (v) contents.

Therefore, canyon users have access to *all* document settings as if they had manually started writing the document from scratch on their own—except they hadn't!—and thus have *full access* to everything Typst in that project.

2 Overview

Out-of-the-box canyon books come with the following:

1. (i) Front-matter; (ii) Body; and (iii) Back-matter book structure;

- (i) Cover page, and (ii) numbered Chapter page images;
- All #set and #show rules are accessible; hence, customizable;
- The “openright” chapter pagination—through #pagebreak(to: "odd") ;
- Front-matter with Title pages; Credits; Contents; List of Figures, Tables, Equations, and Exhibits; and a Preface;
- Book body with regular Chapters (single Part);
- Back-matter with Appendix, Bibliography, Glossary, Index, and Dis-calimers;
- Figures, Tables, Exhibits, Problems, and Equations are Chapter-num-bered: 1. 1 instead of 1, with configurable separators by type *group*, i.e., a separate one for: (i) Equations, (ii) Figures and Tables, (iii) Exhibits, and (iv) Problems;
- A simple and configurable 3-color system with: (i) a thematic color (default: *dark indigo*), (ii) a page background color (default: *light sepia*), and (iii) a cover text color (default: *dark indigo*).

A *selection* of canyon book pages using default settings are grouped in the thumbnail below:

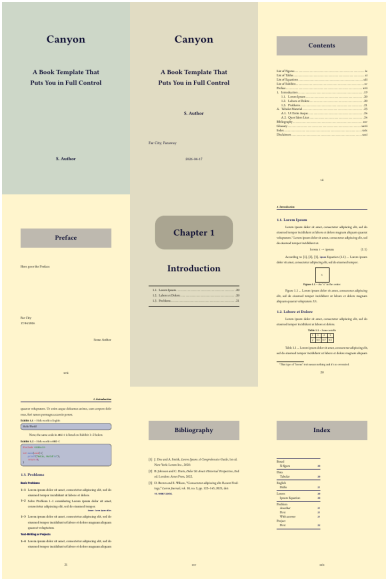


Figure 1: Ordered subset of canyon pages: sample (i) front-matter: half-title, title, contents, and preface; (ii) book body; and (iii) back-matter pages

2.1 Stencil Overview

The stencil tree delivered to the end user upon template instantiation—either through creating a new document from template in `typst.app`, or through the `typst init` command line on a shell—is shown on Listing 1:

```
$ tree -L1F
./
|-- 1-FRONT/
|-- 2-BODY/
|-- 3-BACK/
|-- canyon.typ
|-- RES/
`-- SETUP/
```

Listing 1: Canyon stencil level-1 tree. Entries followed by a slash are directories.

The numbered 1-FRONT, 2-BODY, and 3-BACK directories are for (i) front, (ii) body, and (iii) back-matter book sections—the numbering is to maintain sorting order, specifically on `typst.app` files environment—the RES dir is for document resource files—figures, bibliography, etc.—, and the SETUP dir is for document-wide metadata and settings, as well as for 3rd-party package settings.

3 Suggested Canyon Workflows (Quick Start)

End users are free to adopt any book-writing workflow as they see fit (or even adapt the stencil to something else, as a manual); however, the following ones are suggested as a quick start, for their simplicity, shallow learning curve, and perceived effectivity:

3.1 The Basic Workflow: Metadata and Contents

- Quickly editing `SETUP/META.typ` for document *metadata*;
- Replacing the image placeholders on `SETUP/ELEMENTS.typ` with your actual cover/chapter art `RES/<images>`;
- Being up and running for *content* `2-BODY/<chapters>` creation and adaptation;
- Simply adjust book front- (`1-FRONT/<sections>`) and back- (`3-BACK/<sections>`) matter *contents*.

The scope of the META dictionary defined in `SETUP/META.typ` is the entire book, and includes, as of canyon-0.1.1 the title, subtitle, author, auth, address, description, date, and keywords entries.

This workflow end-result will look like Figure 1 except with your personalized artwork, metadata, contents, and general front- and back-matter portions.

3.2 The Configuration Workflow: General Settings

- All the Basic workflow with:
- Quickly setting *general document configurations* on `SETUP/CONFIG.typ`;
- Editing `__setup.typ` files in the 1- FRONT, 2- BODY, and 3- BACK template stencil directories to more finely control how the corresponding *broad book* sections are displayed.

The scope of the CONFIG dictionary defined in `SETUP/CONFIG.typ` is *main book settings*, and includes, as of canyon-0.1.1 the main lang, paper, size, fonts, colors, and num-seps, i.e., reference chapter-to-item number separators, as the `\u{2013}` in “Problem 1–1”, for instance.

This workflow end-result improves upon the Basic one by setting custom language, paper, book matter sections, text size, fonts, and broad color scheme.

3.3 The Fine-Grained Workflow: Element and Layout

- All the Configuration workflow with:
- Editing `SETUP/ELEMENTS.typ` for fine-grained element *customizations* such as: (i) the set of typst native page, par, text, and block elements; and (ii) custom canyon elemens, such as: exhibit (for code listings, examples, etc.), and problem (for textbooks); or even (iii) creating your own custom element types;

The scope of the ELEM dictionary defined in `SETUP/ELEMENTS.typ` is *various pre-defined book elements*, and includes, as of canyon-0.1.1, native Typst *elements* such as: page, par, text, and block—meaning a *plurality* of *custom* {page, par, text, and block}-setting definitions.

These were designed to be *fully generic*, thus accepting *all valid* named parameter definitions as these native Typst elements can accept—not just a subset thereof—owing to the way this information is used in conjunction with Typst’s spreading `(. .)` operator on dictionaries as illustrated on Listing 2.

This workflow end-result improves upon the Configuration by allowing for a *highly customized book*, than can bear little to no resemblance to the default canyon book style, hopefully achieved without too much pain for the end user.

In `SETUP/ELEMENTS.typ` `canyon-0.1.1` defines, among other variants, the following `ELEM.page` entries as a dictionaries of valid Typst page function named arguments:

```
// blank page
#ELEM.page.insert("blank", (
  paper: CONFIG.paper,
  margin: margins.graphic,
  header: none,
  background: none,
  fill: none,
  numbering: none,
))

// half-title (book cover) page
#ELEM.page.insert("half-title", ELEM.page.blank + (
  background: image(cover, width: 100%),
))
```

This means that `ELEM.page.half-title` “inherits” (builds upon) `ELEM.page.blank` with a overridden (custom) background setting.

This definition allows it to be used as:

```
#page(..ELEM.page.half-title)[
  ...
]
```

in `1-FRONT/half-title.typ` for actually producing the half-title page.

Canyon element definitions are programatically *expressive*—as `half-title` page is just the blank page with a custom background—while maintaining full flexibility—as nothing prevents an end user to add any or all of the valid Typst page named arguments to it, such as binding or columns, etc., according to Typst’s documentation.

Listing 2: Sample canyon element definition and usage design, highlighting its expressiveness and full versatility

4 Citing

This package can be cited with the following bibliography database entry:

```
canyon-package:  
  type: web  
  author: Naaktgeboren, C.  
  title: "Canyon: A Book Template That Puts You in Full Control"  
  url: https://github.com/cnaak/canyon.typ  
  version: 0.1.1  
  date: 2026-04-17
```